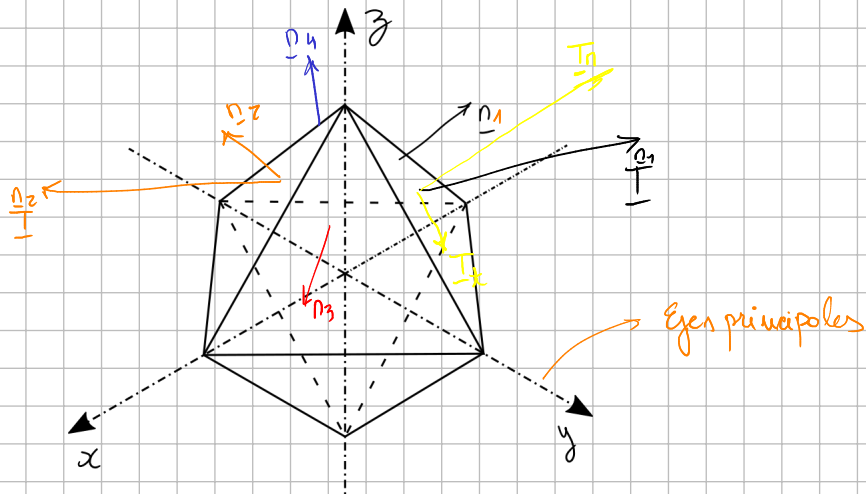




$$\underline{n}^* = \begin{pmatrix} \pm 1 \\ \pm 1 \\ \pm 1 \end{pmatrix} = \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix}$$

$$\underline{n} = \frac{1}{\sqrt{s_1^2 + s_2^2 + s_3^2}} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix}$$

$$\underline{\gamma} = \begin{bmatrix} \gamma_x & 0 & 0 \\ 0 & \gamma_y & 0 \\ 0 & 0 & \gamma_z \end{bmatrix} = \begin{bmatrix} \gamma_1 & 0 & 0 \\ 0 & \gamma_2 & 0 \\ 0 & 0 & \gamma_3 \end{bmatrix}$$

$$\gamma_1 > \gamma_2 > \gamma_3$$

$$\underline{\gamma} = \begin{bmatrix} \gamma_x & 0 & 0 \\ 0 & \gamma_y & 0 \\ 0 & 0 & \gamma_z \end{bmatrix} \Rightarrow \gamma_m = \frac{1}{3}(\gamma_x + \gamma_y + \gamma_z)$$

$$\underline{T} = \underline{\gamma} \cdot \underline{n} = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} \gamma_x s_1 \\ \gamma_y s_2 \\ \gamma_z s_3 \end{bmatrix}$$

$$\begin{aligned} \underline{T}_n = \underline{T} \cdot \underline{n} &= \frac{1}{\sqrt{3}} \begin{bmatrix} \gamma_x s_1 \\ \gamma_y s_2 \\ \gamma_z s_3 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \frac{1}{3} (\gamma_x \frac{1}{s_1^2} + \gamma_y \frac{1}{s_2^2} + \gamma_z \frac{1}{s_3^2}) \\ &= \frac{1}{3} (\gamma_x + \gamma_y + \gamma_z) = \gamma_m \end{aligned}$$

$$\text{Tensor hydrostatisch} \Rightarrow \underline{\gamma}_H = \gamma_m \cdot \underline{I}$$

$$\text{Tensor deviatorisch} \Rightarrow \underline{\gamma}_D = \underline{\gamma} - \underline{\gamma}_H = \underline{\gamma} - \gamma_m \underline{I}$$

$$\underline{T}_k = \underline{T} - \underline{T}_n \cdot \underline{n} = \frac{1}{\sqrt{3}} \begin{bmatrix} \gamma_x s_1 \\ \gamma_y s_2 \\ \gamma_z s_3 \end{bmatrix} - \gamma_m \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} (\gamma_x - \gamma_m) s_1 \\ (\gamma_y - \gamma_m) s_2 \\ (\gamma_z - \gamma_m) s_3 \end{bmatrix}$$

$$\|T_x\| = \frac{1}{\sqrt{3}} \sqrt{(x-x_m)^2 + (y-y_m)^2 + (z-z_m)^2}$$

$$\hookrightarrow y_m = \frac{1}{3}(x+y+z)$$

$$\|T_x\| = \frac{1}{3} \sqrt{(x-y)^2 + (x-z)^2 + (y-z)^2}$$